

Central and Peripheral Nervous System: Neurophysiology

Integrated Biomedical and Clinical Sciences OMFS 512

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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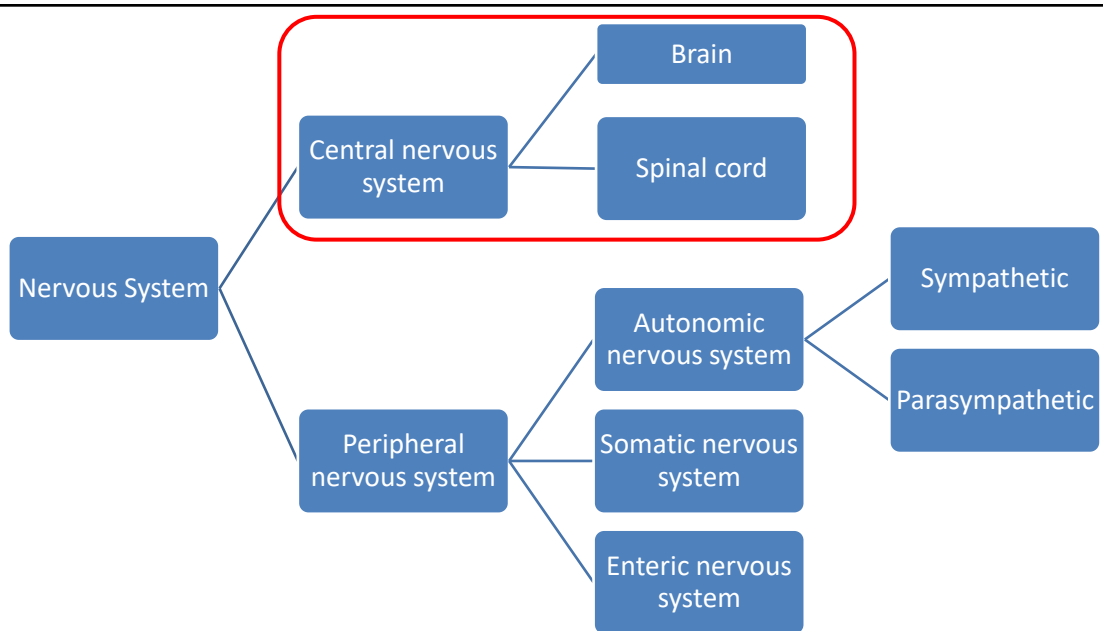
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Learning Objectives

1. Describe components of the central and peripheral nervous system.
2. Describe protective layers of the central nervous system and associated pathologies
3. Describe blood supply to the brain and Compare the different cerebrovascular events.
4. Describe motor and sensory pathways

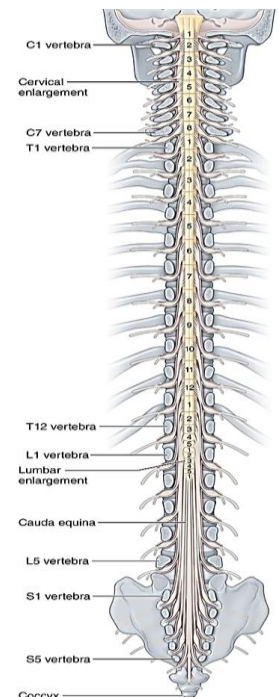
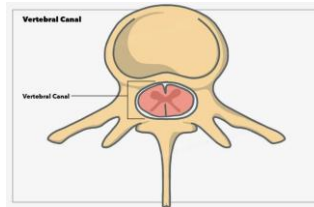
Outline

- Central nervous system:
 - Spinal cord.
 - Protective barriers to the CNS
 - Blood supply and Blood brain barrier
- Peripheral nervous system
- Motor and Sensory pathways (taste).
- Meningitis
- Cerebrovascular events.



Spinal cord

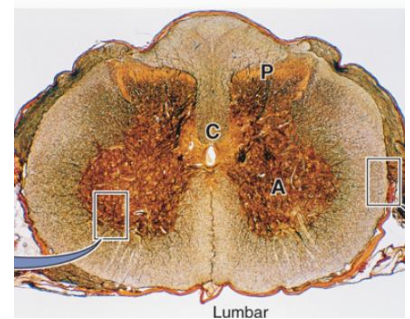
- Extends within the vertebral canal. It occupies around 2 thirds of the vertebral column.
- From the foramen magnum at the cervical region to L1/L2 vertebrae.
- Followed by the cauda equina (lumbosacral nerve roots travelling inferiorly) through the vertebral canal.



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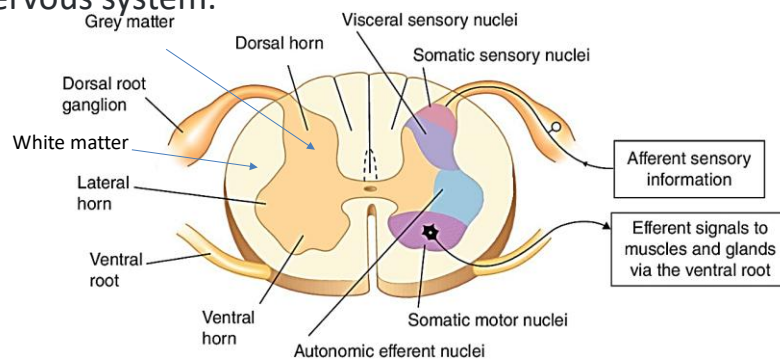
Histology

- White matter is peripheral, and H-shaped grey matter is inside. Which one is myelinated?
- The outer white matter contains myelinated ascending and descending fiber tracts of sensory and motor pathways, respectively.
- The gray matter surrounds a small central canal (c) containing CSF.



The inner grey matter contains nerve cell bodies arranged in horns:

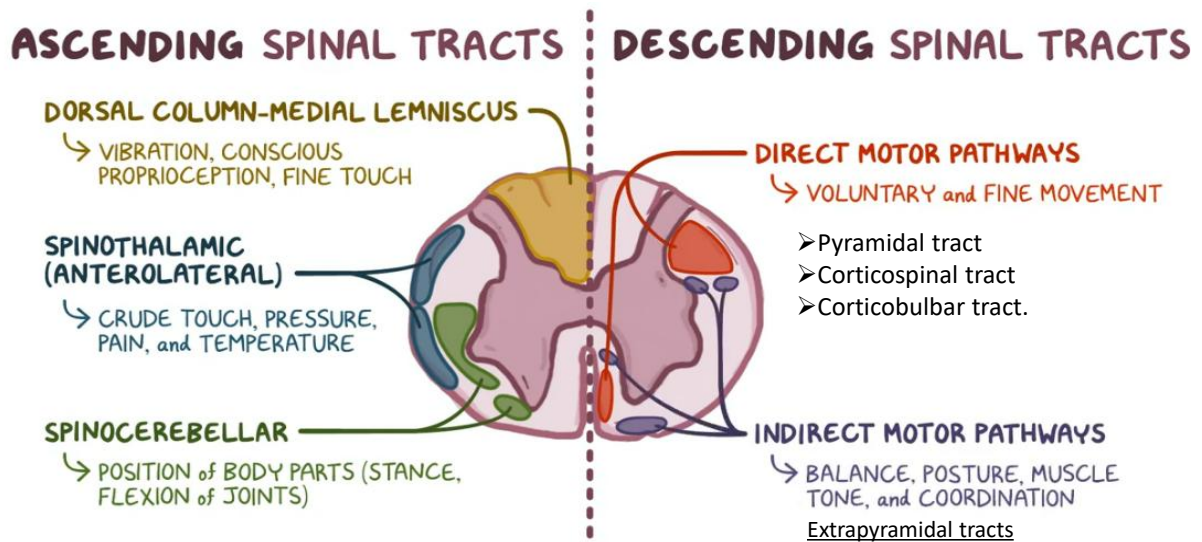
- The anterior (aka, ventral) horn conveys information from the motor cortex to the skeletal muscles (voluntary).
- The posterior (dorsal) horn conveys sensory information to the sensory cortex of the brain.
- The lateral horn contains the cell bodies of preganglionic neurons of the autonomic nervous system.



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White matter Tracts in Spinal Cords



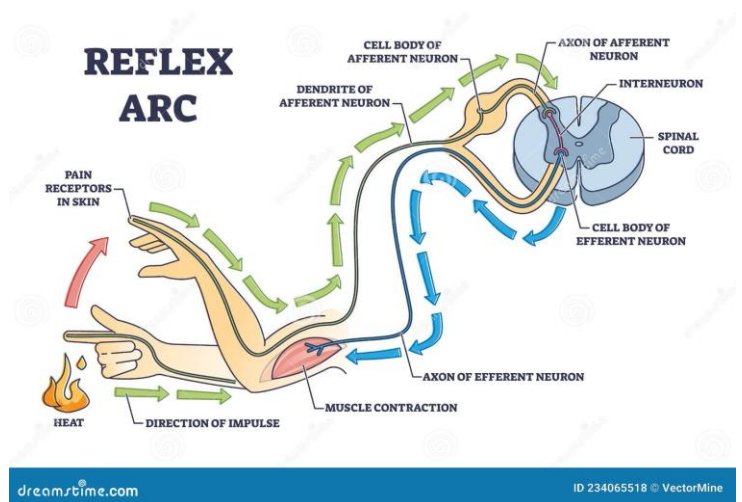
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Functions

- Process sensory input and initiate the reflex responses, either involuntary reflex (e.g. pain withdrawal, corneal reflex) or acquired (e.g. acquired after training like sequence of driving car).
- Conduct sensory input and motor output to and from the brain.
 - Spinal cord in cervical region maintain diaphragm and upper body motions.
 - Thoracic for trunk
 - Lumber region for lower limbs.

Reflex Arc

- Involuntary rapid reflex mediated by the spinal cord before stimulating the brain.
- Components: Receptor + sensory neuron with one or more synapses to generate quick efferent output (via motor neuron + muscle)



Clinical application

Spinal Cord Injuries

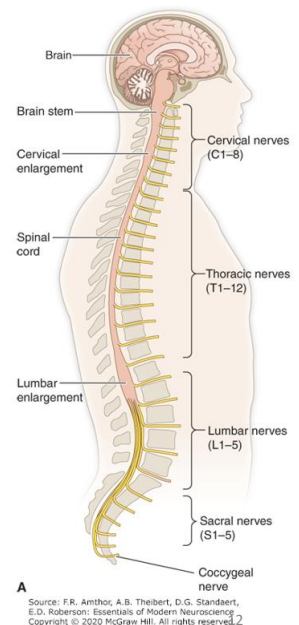
– Cervical region:

- Function: controls sensation/movement neck & upper body. Autonomic function (breathing/HR)
- Transection: C1-C3: no function below the head and ventilator needed. C4-C5: Quadriplegia; C6-C8: loss of lower limb function but some upper limb may be intact.

– Thoracic/Lumbar region:

- Function: Control sensation/movement for upper back, abdominal area and limbs
- Transection: T1-T9 Paraplegia: paralysis in lower body and legs but upper limb intact. T10-L3: some dysfunction in lower limb.

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Protective layers for the central nervous system

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Bone

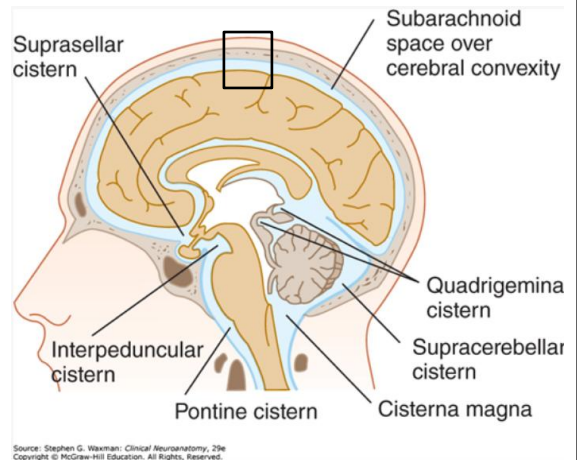
The skull and vertebral column.

Cerebrospinal fluid

Provide hydromechanical protection.

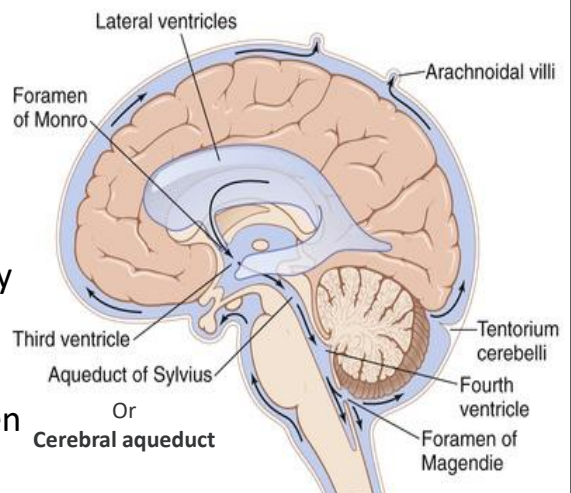
Meninges

Three layers: pia mater, arachnoid and dura mater.



Cerebrospinal fluid

- Clear, colorless and sterile ultrafiltrate of blood.
- Around 150 ml of CSF circulates within the subarachnoid space and ventricles.
- Produced by the ependymal cells, moved by cilia motion and returns to the venous circulation via the arachnoid villi.
- Arachnoid villi act as one-way valve between the subarachnoid space and the venous system.



Composition:

Contains water, ions, glucose, small amounts of proteins and small number of lymphocytes.

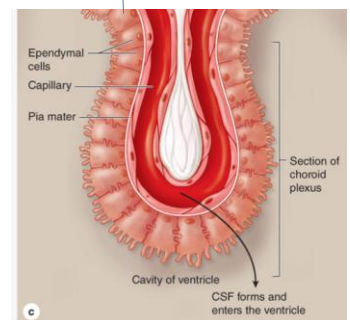
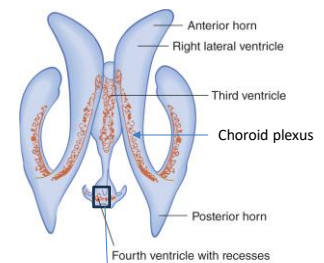
CSF can be sampled via a lumbar puncture (aka, spinal tap) to test composition, clarity and CSF pressure.

Functions:

- a) Buoyancy or hydromechanical protection to brain and spinal cord.
- b) Providing nutrients and removing metabolic waste product.
- c) Aid in neurotransmitters and neuromodulators transport.

Choroid plexus

- Highly vascular tissue (network of capillaries lined by ependymal cells). Projects into the ventricles.
- Main function: remove water/solutes from blood and release it as the CSF.
- Through tight junctions, ependymal cells from blood-CSF barrier & produce around 500 ml of CSF per day.
- Constant secretion of CSF ensures the renewal of CSF renewal 4-5 times/day.
- Reduction of CSF secretion in aging may contribute to neurodegenerative diseases and accumulation of metabolites.



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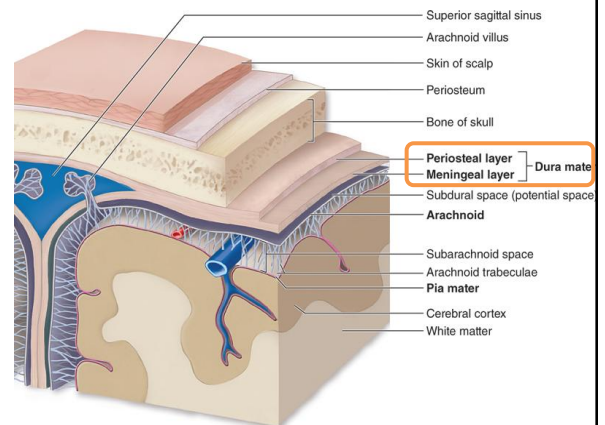
Brain meninges

Dura mater

Outermost layer. Formed by periosteal layer (lining the skull) and an inner meningeal layer (dense collagenous tissue).

Dura is separated from the arachnoid by a narrow subdural space. Subdural space has scarce amount of CSF.

Dural sinuses are cavities between the meningeal and periosteal layers of the dura. Function to drain the deoxygenated blood from the brain to the venous circulation.



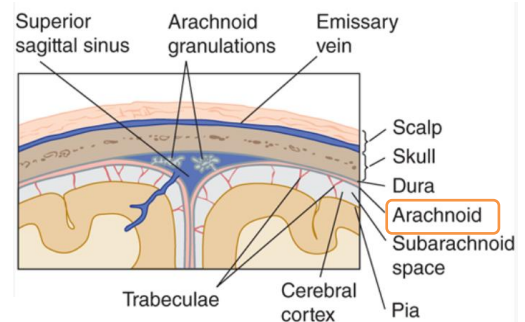
Jonny L. Mescher:
Basic Histology: Text and Atlas, 17th Edition
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Arachnoid mater

A thin, transparent layer that covers the subarachnoid space. Surrounds the brain and spinal cord.

Subarachnoid space: located between the arachnoid and pia mater, and it contains the CSF, blood vessels and cranial nerves.

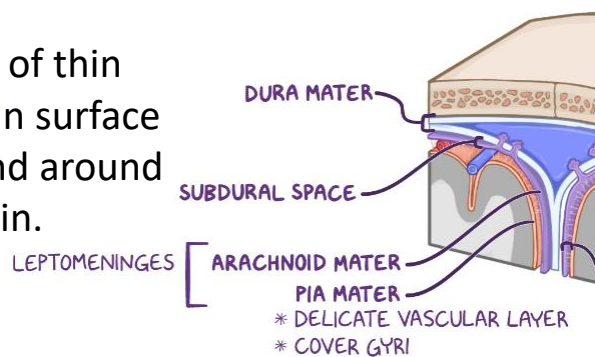
CSF uses the subarachnoid space for circulation and as a conduit to return to the venous circulation via the arachnoid villi (aka, arachnoid granulations).



Citation: Chapter 11 Ventricles and Coverings of the Brain, Waxman SG. *Clinical Neuroanatomy*, 29e; 2020.
Available at:
<https://accessmedicine.mhmedical.com/content.aspx?sectionid=242763936&bookid=28508&resultclick=2>
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Pia mater:

Innermost layer. Composed of thin connective tissue. covers the brain surface and extends into sulci, fissures and around blood vessels throughout the brain.



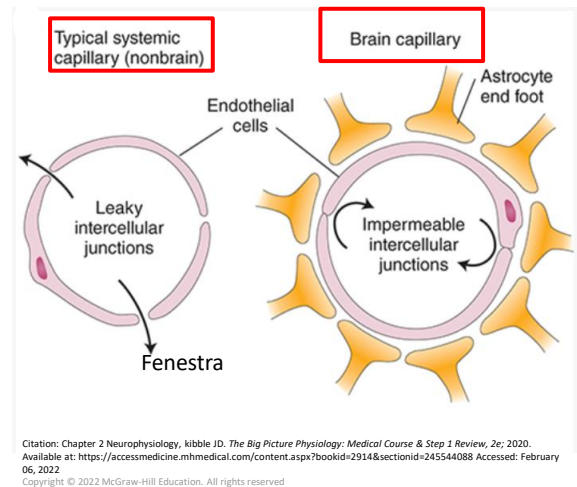
Arachnoid & Pia are named “leptomeninges”.

Meningitis is an inflammatory condition that typically affects the leptomeninges.

Blood brain barrier

The blood brain barrier is a functional barrier made up from:

- 1- The capillary endothelium with continuous tight junctions that seal the seams between endothelial cells.
- 2- Astrocyte end foot strengthening the capillary endothelium.
- 3- Basement membrane.



Function of the blood brain barrier:

1. Tightly controls substances that pass from the blood to the brain.

For example, cholesterol and most proteins can not pass through the blood brain barrier because of their large size. Nonionized (lipid-soluble) drugs can cross the BBB easier than the ionized (water-soluble) drugs.

2. Protects neurons and glia from bacterial toxins, infectious agents, and other exogenous substances. Lipid-soluble substances can diffuse through the capillary endothelial cell membranes (carbon dioxide, oxygen, urea, and ethanol).

3. Prevents the escape of neurotransmitters from their site of production in the CNS to the general circulation.

Inflammation and certain pathological conditions may increase the permeability of the BBB and destroy its selectivity.

Meningitis

Cerebrovascular Diseases

[Cerebrovascular accidents (CVA) or Stroke]

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Meningitis

An inflammatory condition affecting the leptomeninges. In the USA, bacterial meningitis affect 1.4-2.0 per 10⁵ population annually.

- Acute: viral or bacterial infections.
- Chronic: fungal infections.

- **Bacterial meningitis:** *Streptococcus pneumoniae* (most common) and *Neisseria meningitidis* (common); *Mycobacterial tuberculosis* or *S. aureus*.
- **Viral meningitis:** Enterovirus (coxsackievirus and echovirus) and herpes simplex virus.
- **Fungal meningitis:** *cryptococcus sp* in immunocompromised patients.

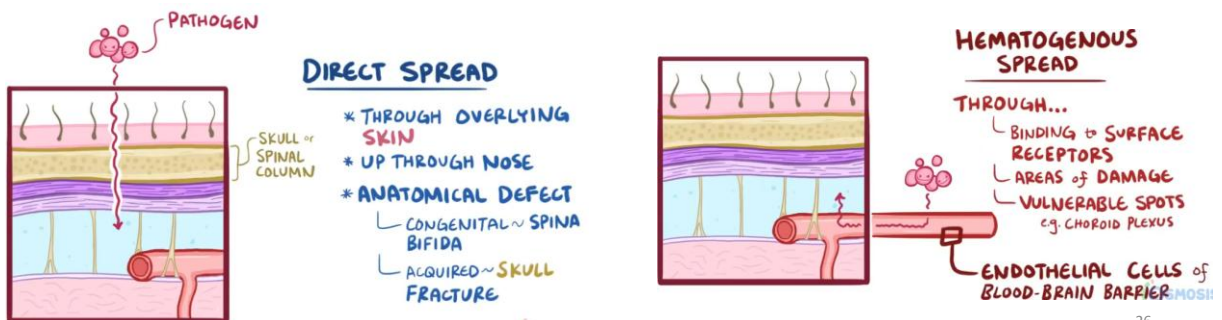
Risk Factors

- Exposure: travel to endemic area or vectors
- Age (infants and elderly)
- Vaccine refusal (bacterial meningitis)
- Immunosuppressed states
- Medical/health conditions (malignancy, VP shunt, HIV, splenectomy, IV drug use, renal failure..).

Pathophysiology:

1- Direct: pathogens can gain access to the leptomeninges through the skin or through the nose. This type can be associated with anatomical defects or skull fractures.

2- Hematogenous: Bacteria colonize the nasopharynx by attaching to nasopharyngeal epithelial cells. Gradually, bacteria can gain access to the bloodstream and reach the intraventricular choroid plexus.

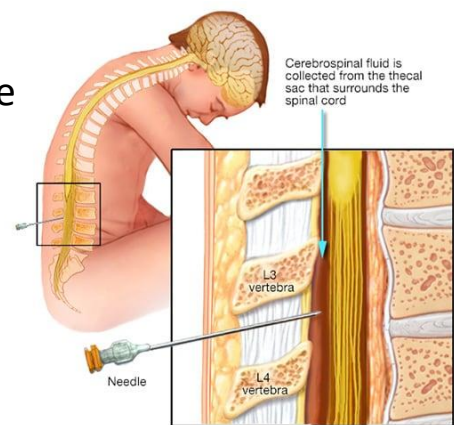
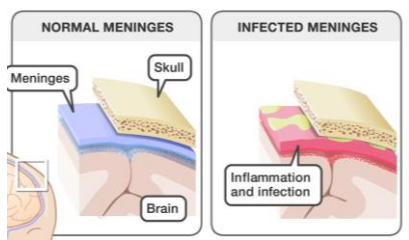


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Diagnosis:

Clinical symptoms. Lumbar puncture to check for CSF clarity, CSF pressure, culture, CSF components (WBCs or protein).



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Symptoms:

- Classic symptoms fever, headaches, and neck stiffness (nuchal rigidity).
- In some cases, photophobia or phonophobia

Complications:

Death, seizures, hydrocephalus, hearing loss..

Prevention:

- Vaccines are effective to prevent bacterial meningitis. Protective vaccines include: meningococcal vaccines (*N. meningitidis*), Hib vaccine (*H. influenzae B*) and pneumococcal vaccine (*S.pneumoniae*).

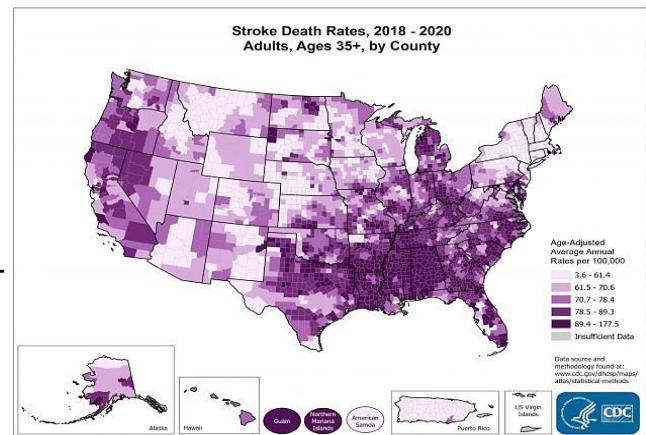
Treatment:

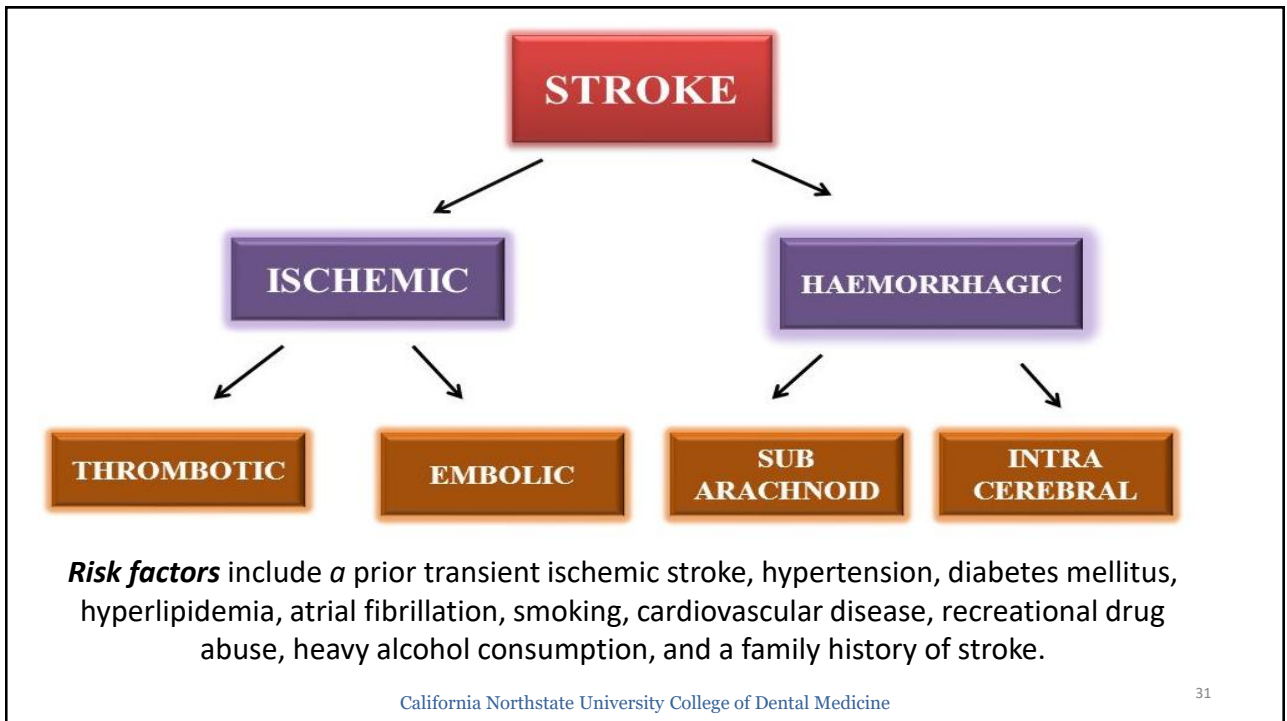
- According to the causative pathogen:
 - Bacterial: treated with dexamethasone (steroid to prevent leptomeninges injury) prior to antibiotics course.
 - Others: specific agent according to the causative agent and supportive.

Cerebrovascular accidents (CVA)

Stroke is the fourth leading cause of death and major cause of disability. Every year > 795,000 people in the US have a stroke.

The risk of having a stroke varies with race and ethnicity (non-Hispanic Black adults have higher risk). The incidence of stroke tends to increase with age (>65Y).





Ischemic stroke

- Prevalence around 85 % of stroke load.
- Caused by a reduced or complete interruption of blood flow leading to rapid depletion of ATP stores. Irreversible nerve damage after 5 min of ischemia. Finally, leading to neural death via necrosis.
- Stroke can affect one side of the brain (unilateral stroke, more common) or both sides of the brain (bilateral stroke , less common).



Risk Factors



TOBACCO



DIABETES



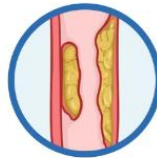
HYPERLIPIDEMIA



HYPERTENSION



**CARDIOVASCULAR
CONDITIONS**



**CAROTID ARTERY
DISEASE**



**ATRIAL
FIBRILLATION**

& Hypercoagulable Disease

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Classified according to the cause:

Thrombotic stroke

Due to thrombus formation inside the cerebral vessels leading to the occlusion of cerebral vessels (either large or small cerebral vessels).

- Symptoms depend on the affected brain area, but they may include headache, dizziness, confusion, loss of coordination, limb paralysis and slurred speech. Can be of sudden or gradual onset of focal deficits. Often preceded by TIAs.

Embolic stroke

Due to arterial occlusion caused by a clot that originates somewhere in the body causing ischemia. Not local, therefore identifying and treating the underlying cause of embolism is essential.

- Symptoms: Difficulty speaking, weakness or numbness, temporary paralysis, blurred vision, nausea, slurred speech. Often with sudden onset of symptoms.

Systemic hypoperfusion.

Refers to problems in the systemic circulation that causes reduced blood flow to the brain as well as other organs. The reduction in perfusion can be caused by arrhythmias, reduced cardiac output, or bleeding.

Symptoms



WEAKNESS



NUMBNESS



INCOORDINATION

Ataxia



FACIAL DROOP



**SLURRED
SPEECH**



**LANGUAGE
DIFFICULTIES**

Aphasia



**CHANGES
in VISION**

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Management:

- 1- Acute phase: thrombolysis, and endovascular surgery.
- 2- Chronic phase: physical therapy to minimize disability and pharmacological treatment to prevent recurrence (e.g: anticoagulant, antihypertensive..)

Impact: long-term disability which can impact the patient's selfcare ability.

If Stroke Happens, Act F.A.S.T.



F—FACE DROOPING

Ask the person to smile. Does one side droop?



A—ARM WEAKNESS

Ask the person to raise both arms. Does one arm drift downward?



S—SPEECH DIFFICULTY

Ask the person to repeat a simple sentence. Are the words slurred?



T—TIME TO CALL 9-1-1

If the person shows any of these signs, call 9-1-1 immediately.

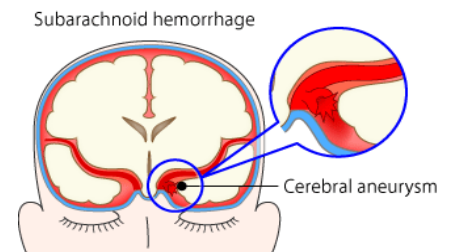
Hemorrhagic stroke

Hemorrhagic stroke, (<20 % prevalence), refers to bleeding in the brain that prevents proper oxygenation leading to edema that potentiates the brain damage.

1- Subarachnoid hemorrhage

Refers to bleeding into the subarachnoid space that surrounds the brain. It is mostly caused by rupturing of an arterial aneurysm that lie at the base of the brain and/or bleeding from vascular malformations.

Symptoms: sudden onset of severe headache (thunderclap headache), blurred vision and loss of consciousness due to sudden rise in intracranial pressure.



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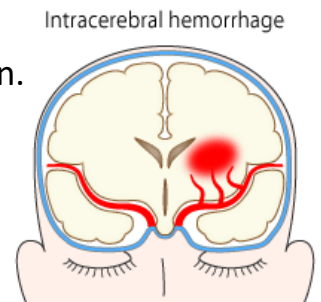
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2- Intracerebral hemorrhage (ICH)

Bleeding directly into the brain parenchyma leading to a localized hematoma that enlarges through blood accumulation. Can be associated with high morbidity and mortality.

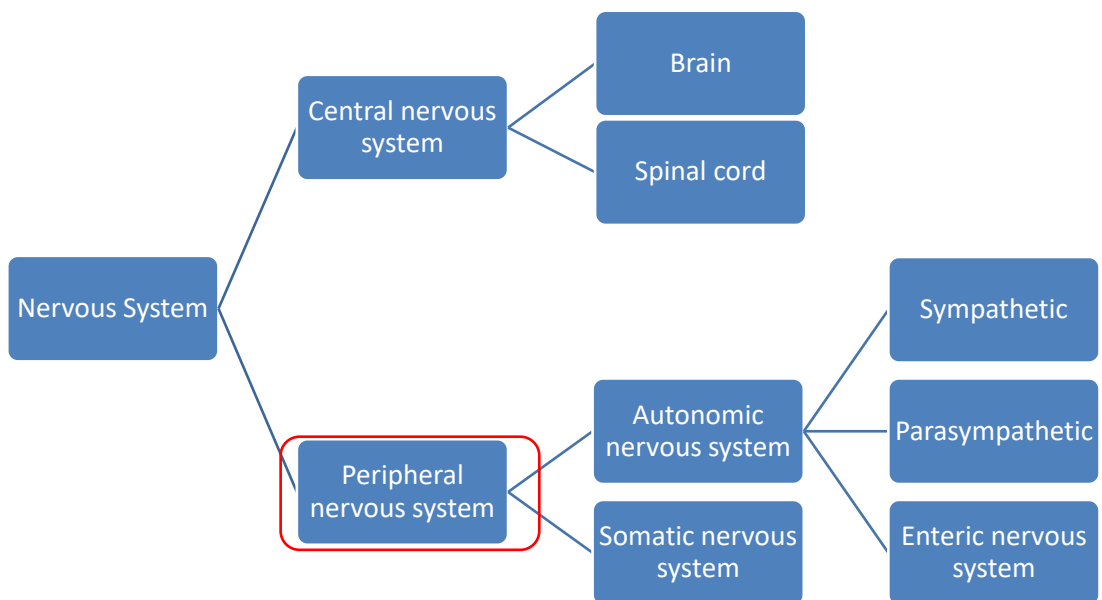
Risk factors include chronic hypertension or hypertensive crisis (Bp>180/120), bleeding disorders, drug use (mostly amphetamines and cocaine), and vascular malformation.

Symptoms: rapid onset (minutes to hours) may include headache, vomiting, decreased level of consciousness and hemiplegia (one sided paralysis). Symptom severity depends on the hematoma size and intracranial pressure.



Transient Ischemic Attack

- TIA is a transient neurologic dysfunction due to brief ischemic event that does not cause acute infarction or tissue injury.
- TIA symptoms are **transient and resolve quickly**.
- The incidence of having a TIA increases with age (> 85 years).
- Risk factors include diabetes, hypertension, age, smoking, obesity, alcoholism, unhealthy diet, psychosocial stress, and previous history of stroke or TIA.
- Symptoms include weakness or numbness of the tongue, face, arm, or leg, trouble speaking normally and double vision or trouble seeing in one or both eyes.

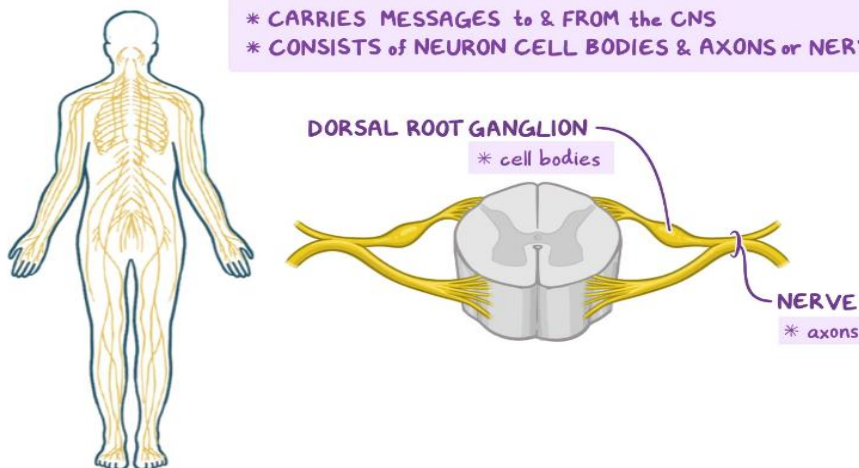


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PERIPHERAL NERVOUS SYSTEM (PNS)

- * CARRIES MESSAGES to & FROM the CNS
- * CONSISTS of NEURON CELL BODIES & AXONS or NERVE FIBERS



Composed of ganglia and nerves. **Ganglia** are clusters of functionally related neuronal cell bodies along with their glial cells. Bundle of axons running together in the PNS is named a **nerve**.

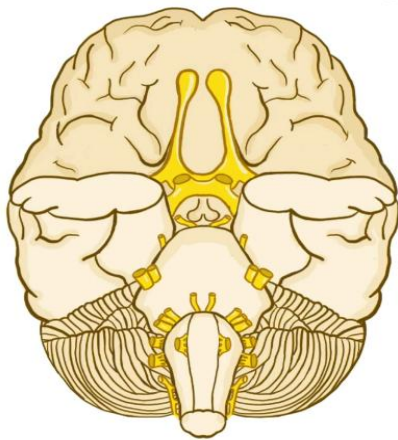
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PNS nerves: Cranial Nerves

~ 12 PAIRS

↳ NUMBERED BASED ON ORDER THEY ARISE FROM NUCLEI IN THE BRAIN
(EXCEPT FOR XI & XII → INVERTED) → EXIT THROUGH FORAMINA



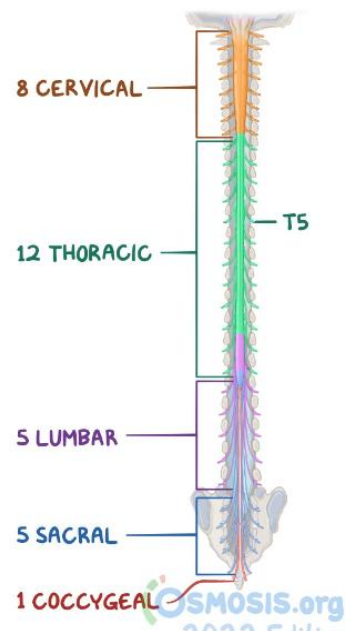
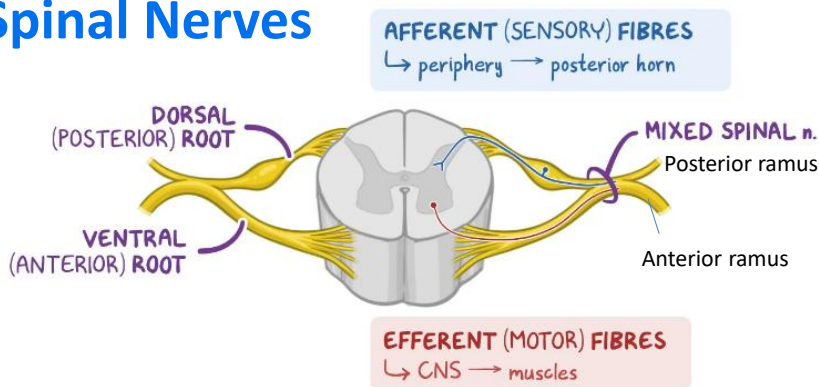
- I - OLFACTORY
- II - OPTIC
- III - OCULOMOTOR
- IV - TROCHLEAR
- V - TRIGEMINAL
- VI - ABDUCENS
- VII - FACIAL
- VIII - VESTIBULOCOCHLEAR
- IX - GLOSSOPHARYNGEAL
- X - VAGUS
- XI - ACCESSORY
- XII - HYPOGLOSSAL

Functionally divided into:

- Sensory nerves
- Motor nerves
- Mixed (sensory & motor; common)



Spinal Nerves

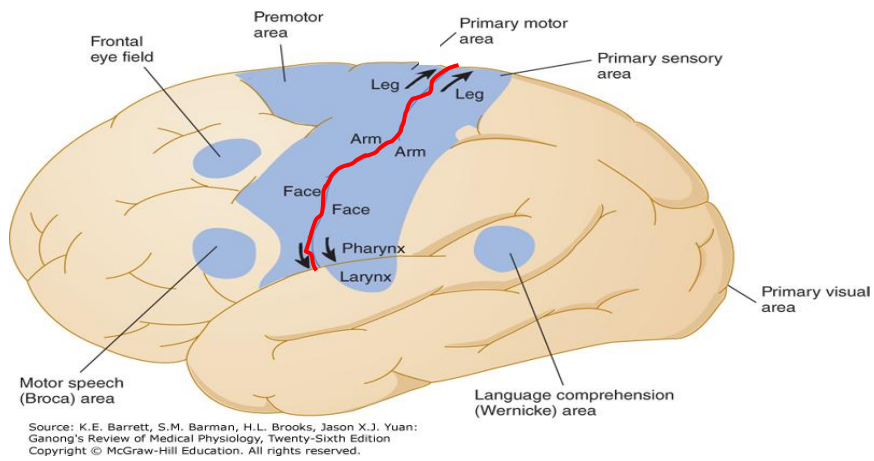


PNS: 31 pair of spinal nerves arise from the spinal cord segments via the dorsal (afferent/sensory) or ventral roots (efferent/motor).

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Motor and Sensory pathways



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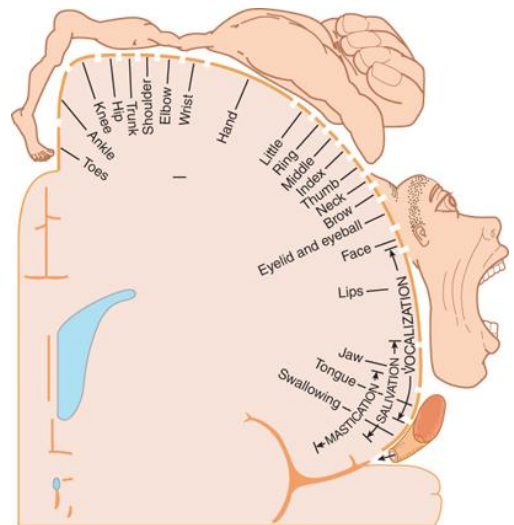
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Motor Pathways

Motor Homunculus

Frontal lobe: Primary motor cortex (coronal section). Sends motor output to body parts.

- Size of the various parts is proportional to the cortical area devoted to them.
- More neurons contributing to the face and hand movement compared to the feet. This enables fine control for face and hands.



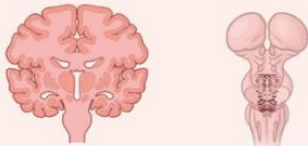
Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition Copyright © McGraw-Hill Education. All rights reserved.

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Descending Somatic Pathways

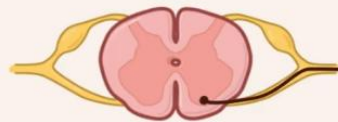
1 UPPER MOTOR NEURONS

~ CEREBRAL CORTEX
~ DEEP NUCLEI of
BRAINSTEM



2 LOWER MOTOR NEURONS

~ ANTERIOR (VENTRAL)
HORNS of SPINAL CORD



INNERVATE
MUSCLES



Carry motor signals from the brain to the muscles to control voluntary movements, like muscle contraction. Five major descending tracts and most of them crossover to the contralateral side (decussate).

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Corticospinal tract

The corticospinal tract controls muscle of the trunk and the limbs.

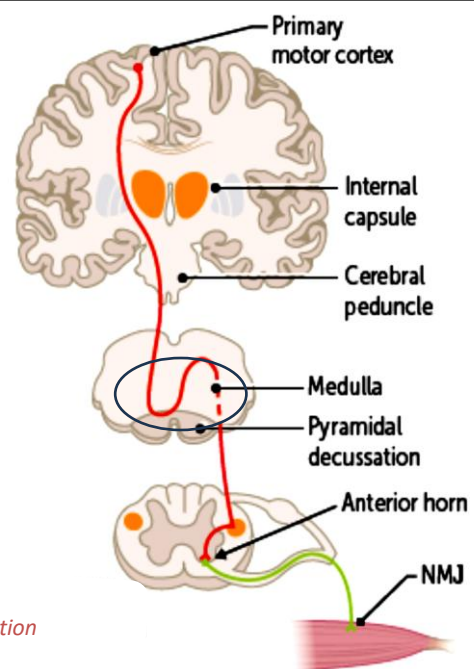
Upper motor neurons:

Cell body are in the motor cortex. Axons descend ipsilaterally and cross over at the medulla to the contralateral side. Then, they synapse at the anterior horn of the spinal cord.

Lower motor neurons:

Transmit the action potential from anterior horn to the muscle.

Therefore, most upper motor neurons injury will cause contralateral dysfunction in the limbs and trunk.



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NMJ= neuromuscular junction⁴⁹

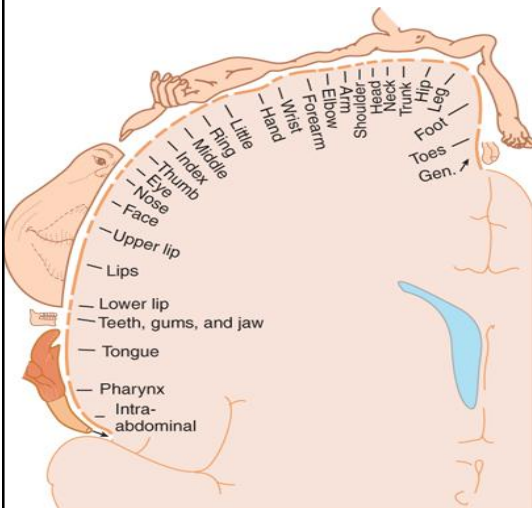
Sensory Pathways

Taste and Pain sensation

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Somatosensory Cortex



Sensory Homunculus

Parietal Lobe: Somatosensory cortex (postcentral gyrus)

- Receives sensory input from different receptors located in the body.
- Size of the cortical receiving area is proportionate to the number of receptors (how sensitive). For example, lips, face and thumb.

Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan:
Ganong's Review of Medical Physiology, Twenty-Sixth Edition
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Sensory Receptors

- Specialized epithelial cells or neurons that can
 - Respond to different environmental signals.
 - Transduce these environmental signals into neural signals.
- Classified according to the stimulus:
 - a. Mechanoreceptors:**
 - For touch and pressure. Examples: free nerve endings in the skin, joint receptors, stretch receptors in muscle, baroreceptors in carotid sinus.

b. Photoreceptors

Rods and cones of the retina.

c. Chemoreceptors

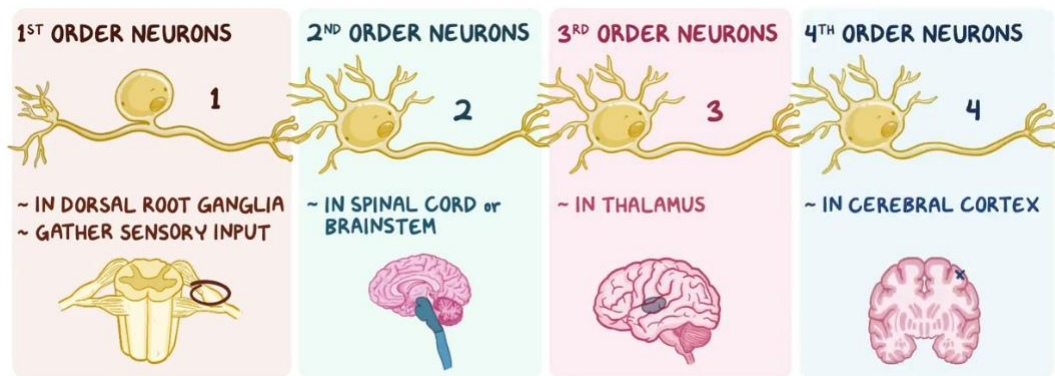
Olfactory receptors, Taste receptors, Carotid body O₂ receptors

d. Nociceptors

Extremes of temperature and pain



Ascending Sensory Tracts



Consists of a relay of 3-4 connected neurons. Most ascending tracts cross over to the opposite side of the CNS (decussate; left cerebral hemisphere receives sensory input from the right side of the body).

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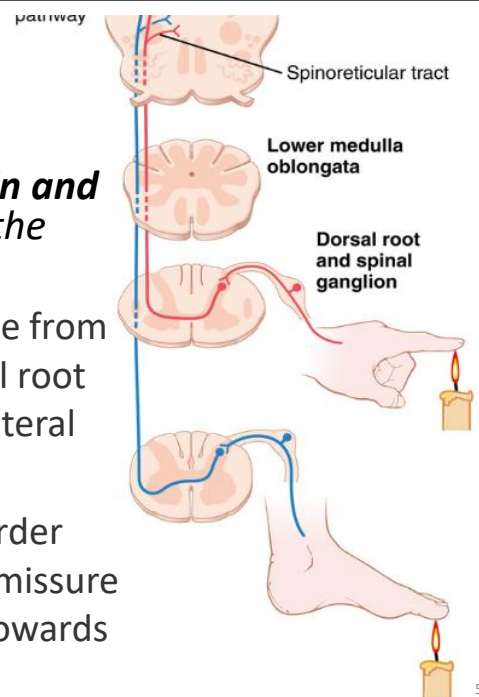
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Spinothalamic tract (anterolateral system)

*Responsible for transferring **sharp pain and heat sensation** from the affected organ to the brain.*

First order neuron: a sensory neuron originate from the skin receptors. Cell bodies are in the dorsal root ganglia in the spinal cord. Synapse at the ipsilateral dorsal horn of spinal cord.

Second order neuron: axons of the second-order neurons decussate via the anterior white commissure of the spinal cord and ascend contralaterally towards thalamus.

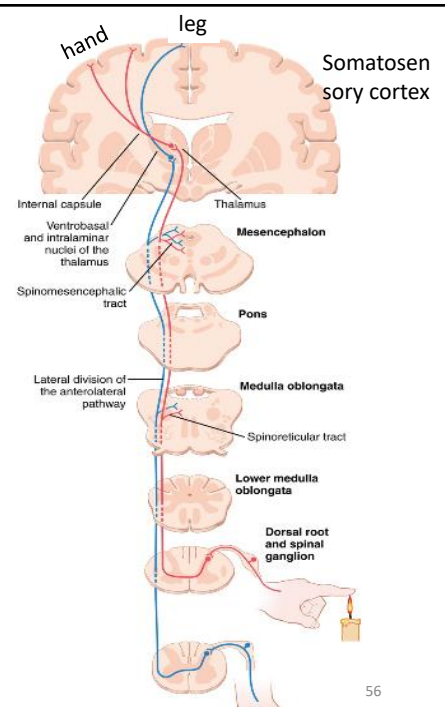


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Second order neuron (cont.): Carry input from the spinal cord to the thalamus.

Third order neuron: carries the information from thalamus to the somatosensory cortex. Cell bodies are in thalamus.

Fourth order neuron: cell bodies are in the appropriate sensory area of the cortex.



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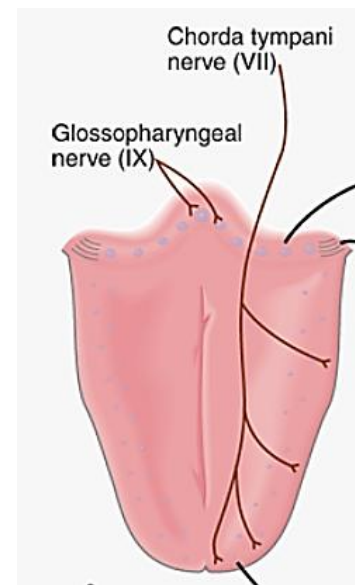
Taste Sensation

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Gustation

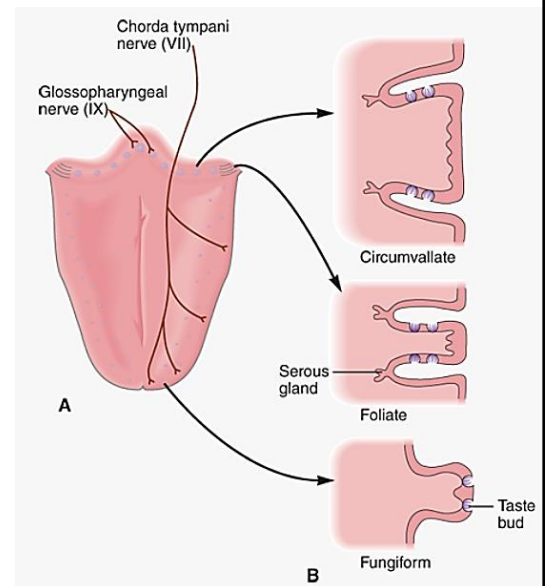
- Taste is sensed by the taste buds in the lingual papillae.
- Taste buds on the anterior two-thirds of the tongue are innervated by the chorda tympani branch of the facial nerve (CN VII)
- Taste buds on the posterior one-third of the tongue are innervated by the lingual branch of the glossopharyngeal nerve (CN IX).



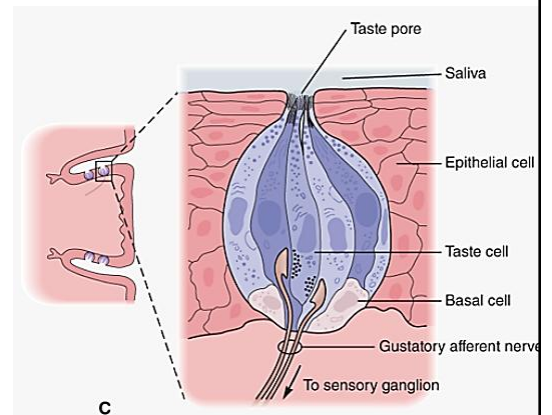
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- Three major types of papillae:
 - **Fungiform:** anterior two-thirds of the tongue. Detects salty, sweet, and umami
 - **Circumvallate and foliate:** posterior third of the tongue. Detects sour and bitter sensations.
- Receptor cells has microvilli to increase the surface area.

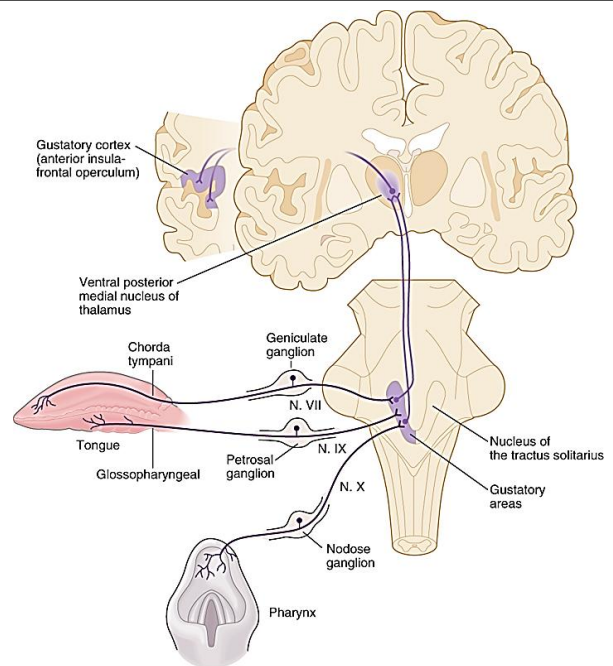


- The taste bud is composed of specialized epithelial cells with microvilli that project into the taste pore.
- Each taste bud contains taste receptor cells and basal cells.
- The basal cells differentiate into new taste cells (taste cells can survive for about 10 days).
- Saliva in the oral cavity acts as a solvent for tastants and help the chemicals diffuses to the taste receptor.



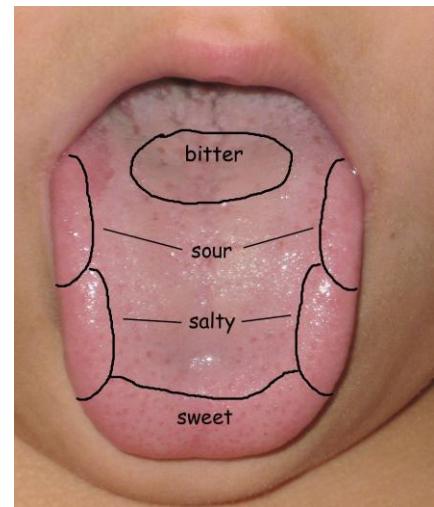
Activation of taste receptors activates neurotransmitter release at the synapse.

Signals travel via the respective afferent cranial nerve relays to the medulla, then thalamus and finally to the gustatory cortex.



California Northstate University | Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition Copyright © McGraw-Hill Education. All rights reserved.

- We can detect 5 general taste categories: salt, sweet, sour, bitter, and umami.
- Taste threshold refers to the minimum concentration at which a substance can be perceived.
- We have the lowest threshold for bitter taste and highest for sweet taste



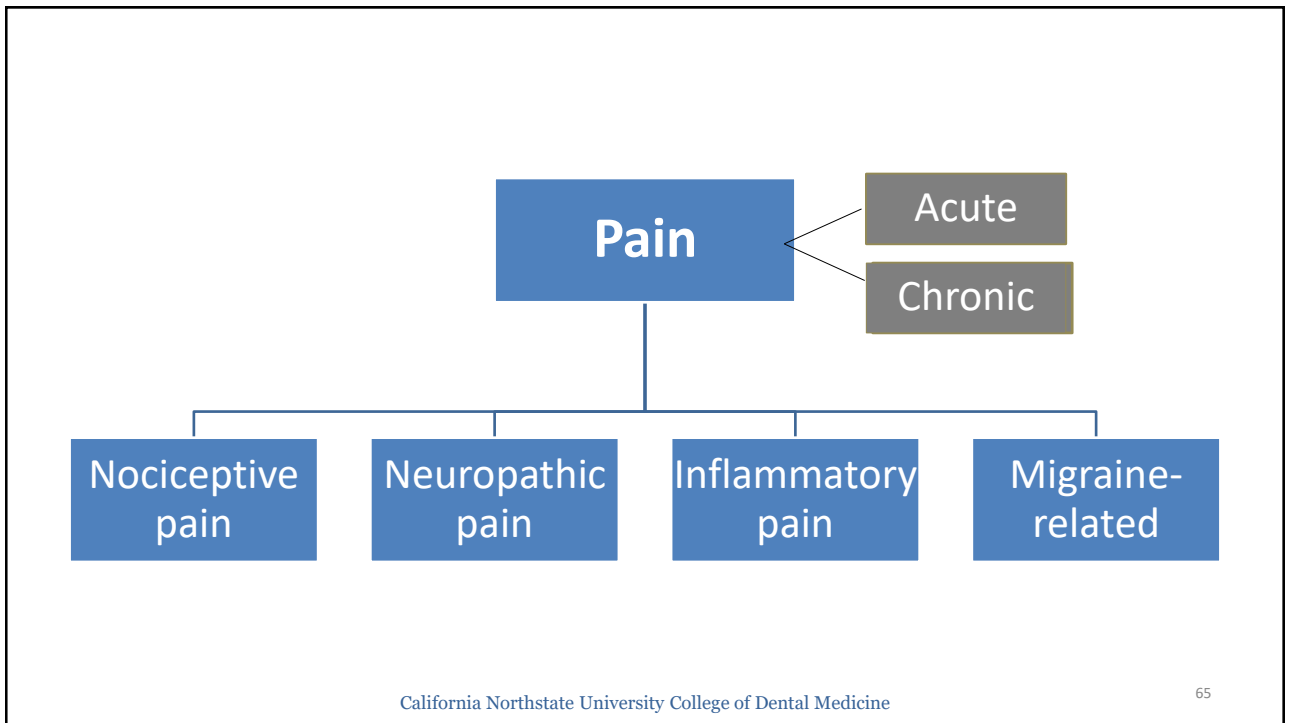
Factors affecting taste sensation

- 1- Integrity of cranial nerves (CN VII, IX & X)
- 2- Taste adaptation: within minutes of eating same food, taste receptors start sending less signals to the brain
- 3- Hunger: increase sensitivity of taste receptors
- 4- Olfactory input.
- 5- Saliva
- 6- Aging: older age decrease the tastebud turnover rate.
- 7- Others: medications, systemic diseases, oral health..

- **Disorders of taste include:**

- ageusia (absence of the sense of taste).
- hypogeusia (diminished taste sensitivity).
- dysgeusia (unpleasant perception of taste).

- They can occur due to nerve damage, several pharmacological agents (captopril, antihypertensive), vitamin deficiencies (like vit. B, zinc), and poor oral hygiene....



Nociceptors and Nociceptive Pain

Nociceptors respond to noxious stimuli by triggering pain to protect the host. Found in skin, dental pulp, joints, some internal organs.

Three main types of nociceptors:

1-Thermal which senses extremely cold or hot temperatures. They are activated by skin temperatures above 45°C or by severe cold (<20°C).

2- Mechanical which can sense excess pressure and deformation (for example, pinch, needle sticks).

3- Polymodal which senses a combination of thermal or mechanical obnoxious stimuli.

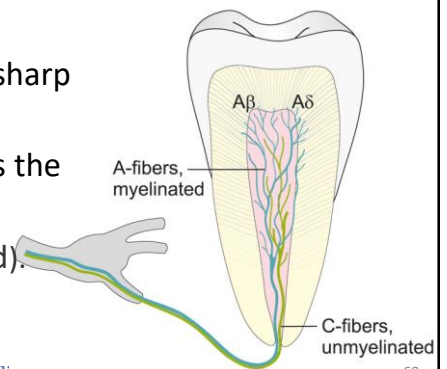
Characters of nociceptors

- 1- Nociceptors are free nerve endings found skin & other tissues.
- 2- Respond to certain threshold of thermal, mechanical and chemical stimuli.
- 3- Non-adaptive nature = unresolved causes, yield continuous pain.
 - Peripheral sensitization: increased receptor responsiveness after injury/inflammation due to mediators (bradykinin, PG, cytokines). Sometimes, continuous stimuli lead to heightened pain (hyperalgesia)
- 4- Pain is affected by rate of tissue damage.
- 5- Slower conduction velocity compared to other sensory receptors.

Fast Pain vs Slow Pain

Afferent neurons associated with nociceptors (A δ & C fibers):

- **A delta** (A δ) fibers are responsible for the fast pain perception.
 - A δ fibers are relatively thin myelinated fibers (1-4 μ m) with fast conduction speed.
 - Transmit localized fast or first pain sensation which is sharp or piercing pain sensation.
 - Activation of A δ fibers releases glutamate that triggers the "first pain".
 - Pain site and intensity can be detected (easily localized).



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- **C fibers** are unmyelinated axons with a smaller diameter (0.5-1 μm) and slow conduction velocity.
 - They are responsible for slow pain perception which is dull, intense, diffuse (poorly localized), aching in nature (like abdominal pain or pain felt in teeth requiring root canal therapy).
 - Slow pain tends to linger after the fast pain subsides.
 - Activation of C fibers releases a combination of glutamate and substance P.
- **A beta ($A\beta$)** fibers are intermediate in size, myelinated. Respond for pressure and touch (**mechanoreceptors**).
- All three nerve fibers are in the dental pulp.

Recommended resources



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